

4.1 Monitors And TVs

The technologies driving monitors and televisions are similar. With the increasing use of PCs as media centres for entertainment purpose, the PC monitor is being replaced with larger displays that were once reserved for entertainment use.

4.1.1 Active Matrix LCD / TFT LCD

In such displays, each liquid crystal in the Liquid Crystal Display is connected to a Thin Film Transistor (TFT) that powers it. This allows the liquid crystals in the matrix to be addressed much faster, allowing for faster response times (this is in contrast to a Passive Matrix LCD).

4.1.2 Aperture Grille

This is an innovation in CRT technology that was developed by Sony for their Trinitron series of TVs. Rather than dots of differently-coloured phosphors, thin horizontal lines of the phosphors are used. A layer of thin metal strips is placed over the lines. These metal strips form the Aperture Grille. The electron beam is allowed by the grille to hit only the correct line of phosphor.

4.1.3 Aspect Ratio

The ratio between the width and height of a display screen. Displays that offer an Aspect Ratio of 4:3 are called “regular” or “standard” displays. Displays that offer an Aspect Ratio of 16:9 or 16:10 are referred to as widescreen displays.

4.1.4 Component Video

This form of video signal transmits the data to reproduce a colour image through three channels : the Luminance channel, which contains information about the image’s brightness / darkness or white / black (called the Y component), the blue



Component video cables